How the Ledger Enforces a Quality Contract Online

Decline is an action; statistics serve cost only; validity is an accounting identity.

1 Step 1: Contract and Plan Menu

Evidence Contract *(Quality Guarantee)*

Maintain the time-average loss (violation rate):

$$\frac{V_t}{t} \le \alpha$$

for all $t \ge 1$

V_t = cumulative loss (violations) up to time t

$\alpha \in (0,1)$ = target quality level

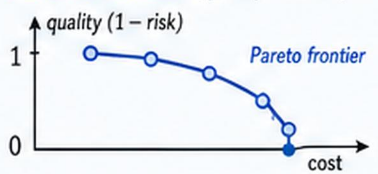
Hybrid RAG Plans *(Action Set)*

Retrieve: BM25, Dense × Generators: LLM A, LLM B, LLM C × Decline Bot: Bot (loss = 0, cost c_{bot})

Each plan $p \in \mathcal{P}$ has cost $c_p > 0$ and risk $r_p \in [0,1]$.

Decline bot: loss = 0, cost c_{bot}, risk $r_{bot} = 0$.

Pareto Frontier *(quality vs. cost)*



Pareto frontier = set of plans not dominated in (cost, risk) space.

Moving along the frontier trades higher cost for lower risk.

Risk Floor *(acceptance threshold)*

0 — $\min_p r_p$ *(risk floor)* — 1 → *risk r*

Contracts (targets α) below the risk floor require the Decline bot.

$\alpha < \min_p r_p \Rightarrow$ only Decline available.

Menu is designed offline and fixed online.
Decline (bot) always available with loss = 0.

2 Step 2: Ledger Gate per Arrival

Incoming Query

q_t at time t

Ledger Reserve *(Quality Budget)*

$$B_t = \alpha t - V_t$$

Depleted — 0 — Healthy

V_t = cumulative loss; αt = allowed loss budget by time t.

Gate Decision

Is $B_t \ge 1 - \alpha$?

NO ($B_t < 1 - \alpha$) → **Force Decline (bot)**

loss = 0

Ledger increases by α (budget accrual):

$V_t \leftarrow V_t$

$B_t \leftarrow B_t + \alpha$

YES ($B_t \ge 1 - \alpha$) → **Sample Action from Action-LP Mixture**

Execute plan $a_t \sim w_t$ and observe outcome.

After Outcome: observe loss $\ell_t \in \{0,1\}$ (1 = violation, 0 = satisfied)

Update Ledger
$V_t \leftarrow V_t + \ell_t$
$B_t \leftarrow B_t + \alpha - \ell_t$

Ledger Shield *(Pathwise Guarantee)*

$V_t \le \alpha t$ for all $t \ge 1$ on every path.

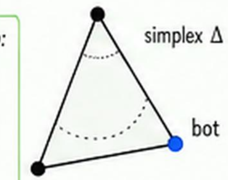
Equivalently, $\frac{V_t}{t} \le \alpha$ for all $t \ge 1$.

3 Step 3: Cost Learning and Audit

Action LP *(learn action mixture w)*

Compute mixture w by solving the linear program:

$$\min_w \; w^\top c \quad \text{s.t.} \; w^\top r \le \alpha, \quad w \in \Delta_{|\mathcal{P}|+1} \text{ (simplex incl. bot)}$$



$c = (c_p, c_{bot})$ = cost vector

$r = (r_p, r_{bot})$ = risk vector, with $r_{bot} = 0$

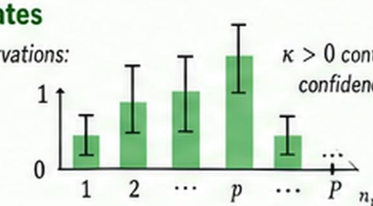
$\Delta_{|\mathcal{P}|+1} = \left\{ w \in \mathbb{R}_+^{|\mathcal{P}|+1} : \sum_i w_i = 1 \right\}$

Optimistic Risk Estimates

For each plan p with n_p observations:

$\hat{r}_p$ = empirical risk$_p$

$\tilde{r}_p = \hat{r}_p + \frac{\kappa}{\sqrt{n_p}}$



$\kappa > 0$ controls confidence.

Two Audit Modes

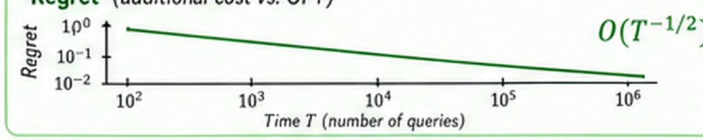
WORST-CASE AUDIT (exact guarantee)

- Uses lower confidence bounds r_p in LP
- Gate based on worst-case risk
- Guarantee: $\frac{V_t}{t} \le \alpha$ holds for all paths

IPW AUDIT (probabilistic)

- Inverse Propensity Weighting for unbiased risk estimates
- Lower variance, cheaper information
- Guarantee holds with high probability

Regret *(additional cost vs. OPT)*



Performance vs. OPT *(Total Cost)*

Ledger (this work) 1.27 – 1.75 × OPT **VS** LTT + decline (baseline) 2.89 – 5.85 × OPT

We learn cheaply, act safely, and audit rigorously—while the ledger guarantees the contract online.